

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

$$5 \left[x^2 - 4x + 4.6 \right]$$

$$5 \left[\left(x + \frac{b}{2} \right)^2 + c - \left(\frac{b}{2} \right)^2 \right]$$

$$5 \left[\left(x + \frac{-4}{2} \right)^2 + 4.6 - \left(\frac{-4}{2} \right)^2 \right]$$

$$5 \left[(x - 2)^2 + 4.6 - 4 \right]$$

$$5 \left[(x - 2)^2 + 0.6 \right]$$

$$\underline{5(x - 2)^2 + 3}$$

$$5(x - 2)^2 + 3$$

(Total for Question 21 is 3 marks)